```python
N = int(input("Enter no. of queens:"))

board = [-1] * N
solutions = []

def is_safe(current_row, test_col):
    for prev_row in range(current_row):
        prev_col = board[prev_row]

        if prev_col == test_col:
            return False

        if abs(prev_row - current_row) == abs(prev_col - test_col):
            return False

    return True

def backtrack_search(row):
    if row == N:
        solutions.append(list(board))
        return

    for col in range(N):
        if is_safe(row, col):
            board[row] = col
            backtrack_search(row + 1)

print(f"--- Backtracking Search: {N}-Queens Problem ---")

backtrack_search(0)

print(f"Total valid solutions found: {len(solutions)}\n")

for idx, sol in enumerate(solutions):
    print(f"Solution {idx + 1}: {sol}")
    for col in sol:
        row_str = ["."] * N
        row_str[col] = "Q"
        print("   " + " ".join(row_str))
    print()
```